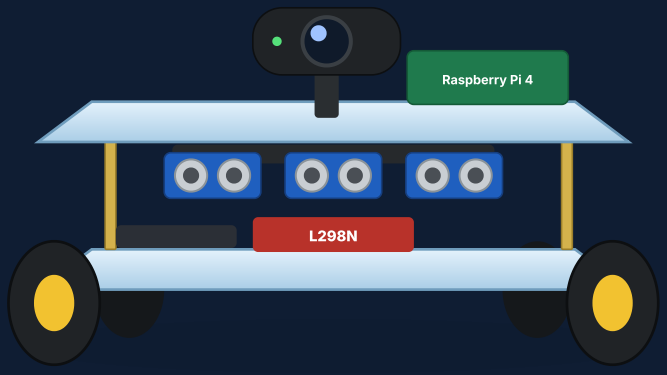
RAQIB · AI INSPECTION ROBOT · RELIABLE BUDGET EDITION

Build the robot

Under RM 500: what to buy, every part pictured and explained, how to wire each sensor, and how to put the robot together, step by step.



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Final-year IoT project · UNISIRAJ

Guide version 2.5 · October 2026
Matches the Raqib software 2.5 · parts ≈ RM 495

Inside this guide

1. **How the robot works** - the parts and how they talk to each other
2. **Shopping list** - every part, how many, why, and an estimated price (under RM 500)
3. **Meet the parts** - a picture of each part and the pins you will use
4. **Safety first** - the battery, voltages and moving wheels
5. **The pin map** - every Raspberry Pi pin the robot uses
6. **Build it in 11 steps** - assembly, wiring and calibration, one part at a time, with a test after each
7. **Troubleshooting** - what to do when something does not answer
8. **Final checklist** - before the first demo
9. **Upgrades later** - parts you can add when you have more budget

> **The golden rule:** wire one part, test it, then add the next. When something does not work, you know it is the part you just added. Every step ends with a check box list and a test command.

> **Wire colours used in every diagram**
>
> 5 V · 3.3 V · GND · SDA · SCL · battery + · motor · PWM · signal
>
> Use the same colours for your real jumper wires where you can: it makes mistakes easy to spot.

> **Practise first, for free:** the Wokwi picture guide lets you wire most of these sensors in a browser simulator, and the phone app's *Try the demo robot* button shows what the finished robot does.

1 How the robot works

The Raspberry Pi 4 is the brain. It runs the robot program and also its own small server, so the phone app talks straight to the robot over Wi-Fi: no laptop. Two batteries keep it reliable: a USB power bank feeds only the Pi, and a 2 × 18650 pack feeds only the motors, so a motor start can never restart the Pi. A hand-made **hub board** shares power and the I²C bus with the sensors, and each sensor group has one job.

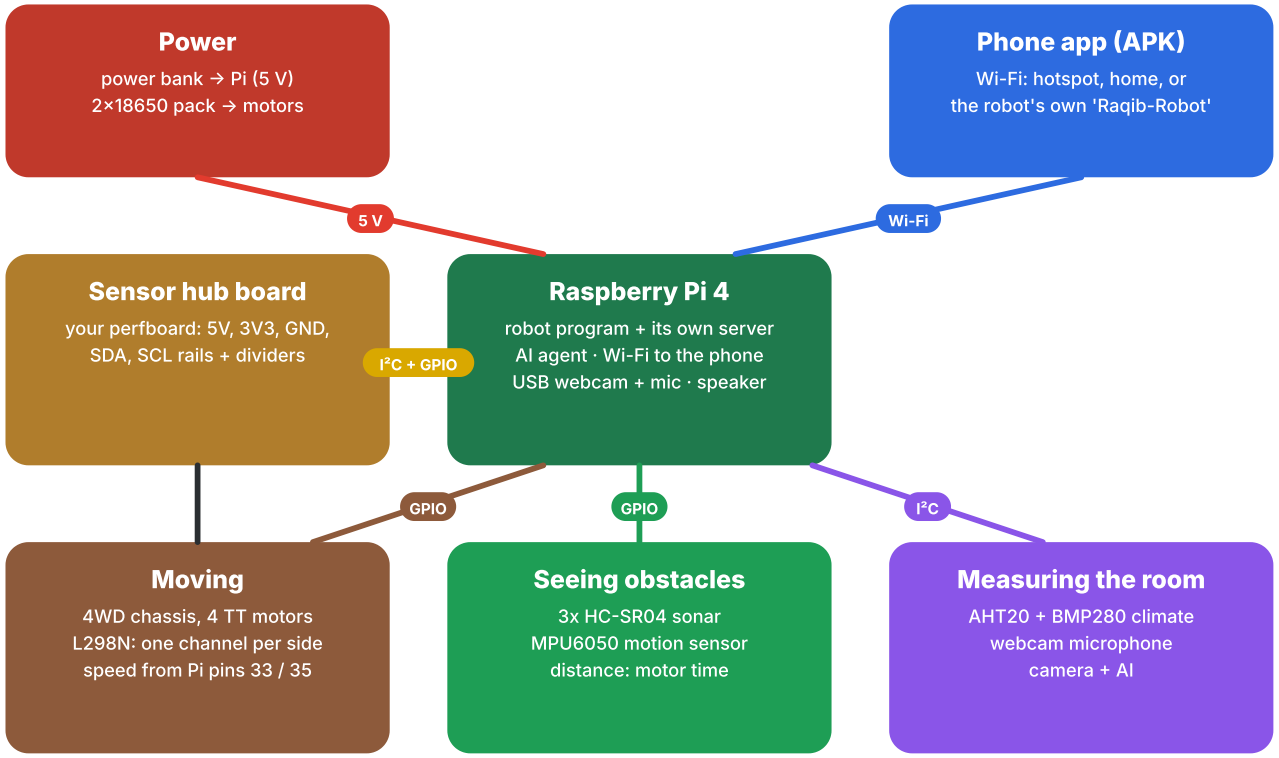


Figure 1 - the five groups of parts around the Raspberry Pi.

Two kinds of connection

I²C bus (2 wires, SDA + SCL): two boards share it, each with its own address - MPU6050 0x68, AHT20 0x38 + BMP280 0x77.

GPIO pins: one wire per signal - motor direction and speed, sonar trigger and echo.

Two voltages

5 V for the sonars and the amplifier.

3.3 V for everything that talks to the Pi. A Pi pin must **never** see 5 V: the voltage dividers on the hub board take care of the sonars' 5 V Echo outputs.

Built to work first time. This build leaves out the parts that most often cause trouble in a first robot: the 5 V converter that must be adjusted by hand, wheel encoders, the gas sensor and its ADC. Distance comes from motor time × a speed you calibrate once (step 10), turns from the MPU6050. All of them can be added later (section 9).

The camera does not move. Instead of servos, the robot turns its whole body to look left or right (the app's *Drive → Camera* pad works the same), and turns back before it drives on. This saves about RM 40.

2 Shopping list

Prices are **estimates in Malaysian ringgit** from typical Malaysian online electronics shops (Cytron, Techmakers, Shopee / Lazada sellers) in 2026. Check current prices and stock before you buy; shop around and the total can drop further.

#	Part	Qty	RM each	RM
The brain				
1	**Raspberry Pi 4 Model B, 2 GB** Runs the robot program and its own server. 2 GB is enough: the AI runs in the cloud	1	240	240
2	**microSD card 16 GB, A1 / Class 10** Holds Raspberry Pi OS and the robot software	1	12	12
3	**Aluminium heatsink set for the Pi 4** Keeps the Pi from slowing down when it gets warm	1	4	4
Power (two separate batteries: the most reliable choice)				
4	**USB power bank 10,000 mAh, 5 V 3 A output, with its USB-C cable** Powers the Pi only. Motor starts can never reset the Pi, and no voltage to adjust	1	40	40
5	**18650 Li-ion cells, flat top, 2500-3000 mAh** Motors only. Known brands; avoid fake '9900 mAh' cells	2	15	30
6	**2 × 18650 battery holder with on/off switch and leads** 7.4 V motor pack; its switch is the motor power switch	1	6	6
7	**2-slot USB 18650 charger** Charges the two motor cells	1	12	12
Moving				
8	**4WD smart car chassis kit: 2 acrylic decks, 4 TT motors (3-6 V), 4 wheels, brass standoffs, screws** The body: four driven wheels (skid steering)	1	40	40
9	**L298N dual H-bridge motor driver** Turns the Pi's small signals into motor power	1	9	9
Sensing				
10	**HC-SR04 ultrasonic distance sensor** Left, centre, right: stops before obstacles and steers round them	3	4	12
11	**MPU6050 motion sensor (GY-521 board)** Knows how far the robot has turned and keeps it straight	1	8	8
12	**AHT20 + BMP280 combo board** Temperature, humidity and air pressure	1	10	10
Camera and sound				
13	**USB webcam 720p with built-in microphone** Photos, live video and hearing in one part	1	30	30
14	**PAM8403 amplifier board** Makes the voice loud enough	1	4	4
15	**Speaker 3 W 4 Ω, about 40 mm** The robot's voice	1	5	5
16	**3.5 mm male audio cable (you will cut one end)** Pi audio jack → amplifier	1	3	3

#	Part	Qty	RM each	RM
Wiring and mounting				
17	**Resistors: 1 kΩ x 3, 2 kΩ x 3** Echo dividers: protect the Pi from the sonars' 5 V	1	3	3
18	**Perfboard / stripboard about 7 × 9 cm** Your 'hub board'	1	3	3
19	**Jumper wires: F-F and M-F, 40 of each** Most connections plug in, no soldering	1	8	8
20	**Male + female pin header strips** Plug-in points on the hub board	1	3	3
21	**22 AWG wire, red + black, 1 m each** Motor and battery wiring	1	5	5
22	**M2.5 screws + nylon washers (the kit already has the deck standoffs)** Raspberry Pi and board mounting	1	4	4
23	**Zip ties, double-sided foam tape, heat-shrink** Tidy, safe, nothing rattles loose	1	4	4
	Total for the robot (estimate, under RM 500)			**≈ 495**

Tools (borrow them from the lab: not counted in the RM 495)

Tool	RM
Soldering iron with solder and stand Motor leads, hub board, header pins	50
Digital multimeter Check every voltage BEFORE connecting a board	40
Small screwdriver set (Phillips + flat) Screw terminals and chassis	15
Wire stripper / side cutter Clean wire ends	15
Hot glue gun Fix cables and the sonars	20
Tape measure (3 m or more) Calibrating the driving distance (step 10) and measuring the rooms	10

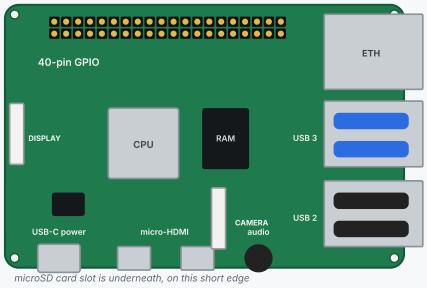
Buying tips

- **Save more:** a used Raspberry Pi 4 (2 GB or 4 GB) often sells for RM 150-200. Ask your department too: many labs lend Pis to final-year students.
- **18650 cells:** buy known brands from a trusted seller. Real cells are 2500-3500 mAh; anything claiming "9900 mAh" is fake and unsafe. Flat-top cells fit most holders.
- Check the **chassis kit** photo: 4 TT motors, 4 wheels, 2 decks and standoffs.
- **Power bank:** it must say **5 V 3 A** (or 15 W or more) on the output. 2 A banks make the Pi show low-voltage warnings. Some banks switch off when little current flows; the Pi draws enough, so this is not a problem.
- **HC-SR04:** buy one spare (RM 4): it is the part that most often arrives dead.
- **Webcam:** any small UVC "plug and play" USB webcam with a built-in microphone works; the robot finds it by itself.
- No 2 kΩ resistors? Two 1 kΩ in series make 2 kΩ.
- Have a phone charger with a USB-C cable? Use it to power the Pi on your desk while you install the software, and save the power bank's charge.

Budget: about **RM 495** for the robot. Nothing in the app is lost: the same software runs this build and the bigger one, and finds by itself which parts you fitted. See *Upgrades later* at the end.

3 Meet the parts

What each part looks like, what it does, which pins you will use and one tip. The pictures are drawings of the common versions; your boards may differ slightly in colour or pin order, so always read the labels printed on the board.

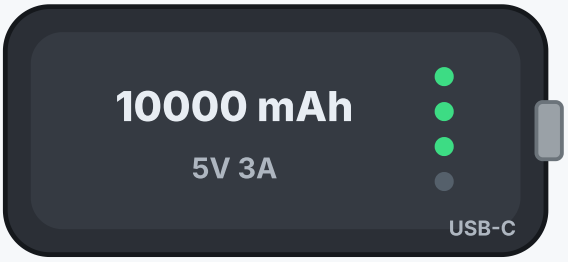


Raspberry Pi 4

The robot's computer: drives, senses, runs its own server for the phone.

Connections: 40-pin GPIO header (top edge) · USB-C power · USB ports for the webcam · 3.5 mm audio jack for the speaker

💡 2 GB is enough. Put the microSD card in the slot underneath on the short edge.

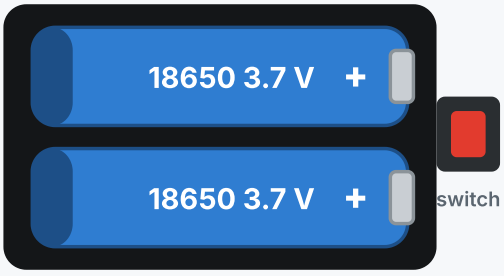


USB power bank (Pi battery)

Powers the Raspberry Pi, and nothing else.

Connections: USB-C (or USB-A) output → its own cable → the Pi's USB-C power socket

💡 Choose one that says 5 V 3 A (15 W) on the output. Its LEDs show how full it is; charge it like a phone.

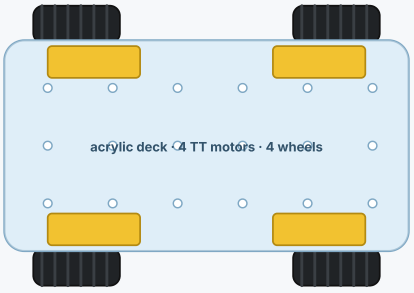


Motor battery: 2 × 18650

Powers the motors only, through the L298N.

Connections: Red + → L298N +12V · black – → L298N GND

💡 7.4 V nominal, 8.4 V full, about 6.6 V empty. Its switch turns the motors on and off.

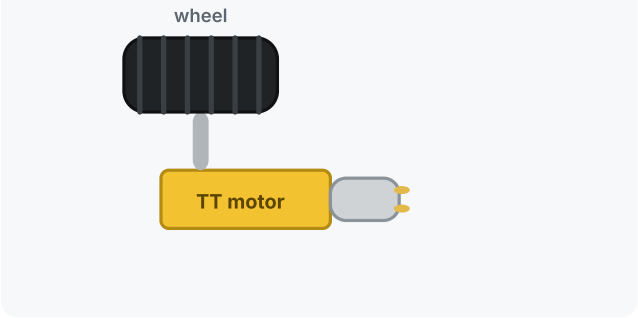


4WD chassis kit

Two acrylic decks, four TT gear motors, four wheels and brass standoffs.

Connections: 4 motors under the bottom deck · second deck on the standoffs

💡 Check all 4 motors and wheels are in the box before you start.

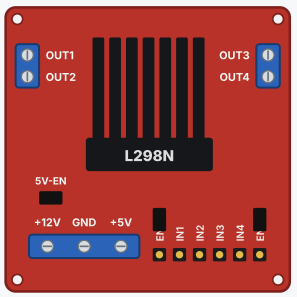


TT gear motor

3-6 V geared motor; two per side, wired together.

Connections: 2 solder tabs per motor (red / black wires)

💡 Solder wires before you mount the motors; it is hard afterwards.

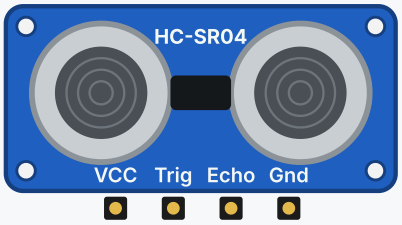


L298N motor driver

Switches motor-battery power to the motors forwards or backwards.

Connections: +12V / GND from the motor battery · GND also → Pi pin 9 · OUT1/OUT2 left pair · OUT3/OUT4 right pair · IN1-IN4, ENA/ENB from the Pi

💡 Remove the ENA and ENB jumpers. Keep the 5V-EN jumper. The GND wire to the Pi is essential.

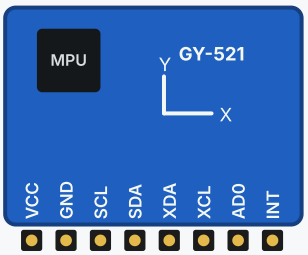


HC-SR04 ultrasonic

Measures distance with sound, 2 cm to 4 m.

Connections: VCC 5V · Trig (from Pi) · Echo (5 V out! needs a divider) · Gnd

💡 Buy 4: one spare. They sometimes arrive dead.

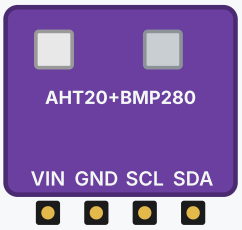


MPU6050 (GY-521)

Gyroscope + accelerometer: the robot's sense of turning.

Connections: VCC · GND · SCL · SDA (the others are not used)

💡 Mount it flat, near the middle of the chassis, with foam tape. Cheap clones work too.



AHT20 + BMP280

Temperature and humidity (AHT20) plus air pressure (BMP280) on one board.

Connections: VIN · GND · SCL · SDA

💡 Keep it away from the Pi: it is warm.

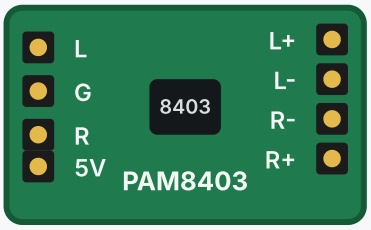


USB webcam with microphone

The robot's eyes and ears: photos, live video and sound.

Connections: Any USB port (a blue USB 3 port is best)

💡 Choose one sold as 'plug and play / UVC' with a built-in microphone.



PAM8403 amplifier

Small 3 W amplifier for the speaker.

Connections: 5V/GND power · L G R input from the Pi's audio jack · L+ L− output

💡 Use only the left channel (L+ / L−).

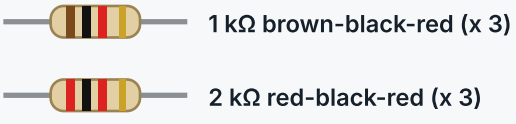


Speaker 3 W 4 Ω

The robot's voice.

Connections: 2 wires to L+ and L−

💡 Face it outwards at the back of the top deck.



Resistors

Three voltage dividers: they turn the sonars' 5 V Echo into a safe 3.3 V.

Connections: No polarity: either way round

💡 No 2 kΩ? Two 1 kΩ in series make 2 kΩ.

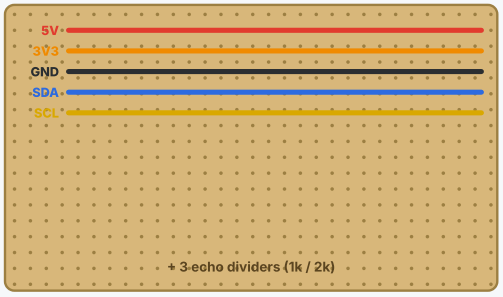


Jumper wires

Plug-in connections.

Connections: F = socket (fits on a pin) · M = pin

💡 Use colours: red 5 V, orange 3.3 V, black GND.

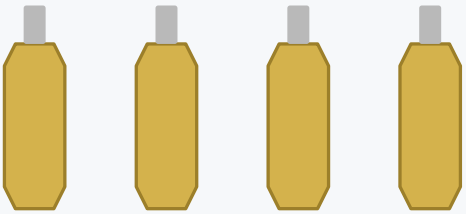


Hub board (you make it)

One place where power, the I²C bus and the sonar dividers meet.

Connections: 5V · 3V3 · GND · SDA · SCL rails + 3 dividers

💡 Step 5 shows how to build it.



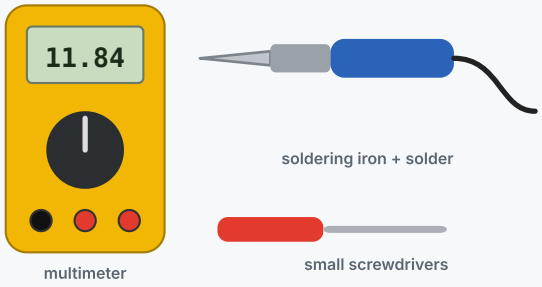
M3 standoffs 35-40 mm + screws

Standoffs and screws

Hold the second deck and the boards.

Connections: M3 for the deck, M2.5 for the Pi

💡 Nylon washers stop shorts under boards.



Tools

What you need on the desk (borrow them from the lab).

Connections: Soldering iron, multimeter, screwdrivers, stripper, hot glue

💡 Measure, then connect. Always.

4 Safety first

Lithium batteries store a lot of energy.

- Never short the + and − of a cell or pack (no loose wires, no metal tools across terminals).
- Check polarity three times before inserting a cell: + goes to the + mark.
- Charge only with a proper Li-ion charger, never unattended, on a non-flammable surface.
- Do not use cells with dents, torn wrapping or that get hot; recycle them.
- Switch the battery off before changing any wire.

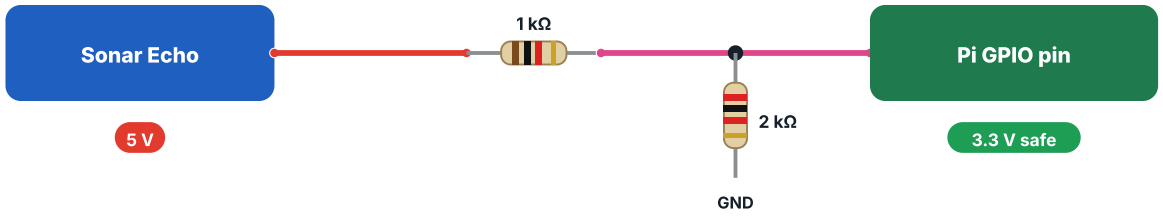
Protect the Raspberry Pi.

- Its GPIO pins are **3.3 V only**. 5 V on a GPIO pin damages it instantly, so every 5 V output goes through a divider.
- The Pi gets power only from the power bank's USB cable. **Never** connect the motor battery's + to the Pi or the hub.
- Never power motors from the Pi's pins. The only wire between the motor side and the Pi side is GND (pin 9).
- Measure each power rail with the multimeter before plugging a board in.
- Touch something metal before handling boards (static electricity).

Moving parts.

- Test motors with the robot on a box, **wheels in the air**.
- Keep fingers and hair away from the wheels.
- The L298N heatsink gets hot.

The two rules you will use most



A voltage divider: 5 V × 2 kΩ / (1 kΩ + 2 kΩ) = 3.33 V, safe for the Pi.

And: **measure, then connect**. Put the multimeter on DC volts, black probe on GND, red probe on the rail. 5 V rail: 4.9-5.2 V. 3.3 V rail: 3.2-3.4 V. Motor battery: 6.6-8.4 V.

5 The pin map

Every Raspberry Pi pin the robot uses. Physical pin numbers (1-40) are what you count on the header; the GPIO names are what the software uses (they are already set in `edge/config.yaml`).



Figure 2 - Raspberry Pi 4 header, top view, USB ports pointing down the page.

> Everything else does not touch the header: the webcam (camera + microphone) uses USB, the speaker amplifier uses the 3.5 mm audio jack, and the Pi is powered through its USB-C socket from the power bank.

6 Build it in 11 steps

Each step lists what you need, what to do, a picture, and a check. Do them in order.

1 Prepare the Raspberry Pi (software first)

You need: Pi 4, microSD card, the power bank, your PC

1. On the PC install **Raspberry Pi Imager**. Choose *Raspberry Pi 4 → Raspberry Pi OS (64-bit)* (the Lite version is fine) → your microSD card.
2. In the settings (gear / "Edit settings"): hostname `robot01`, a username and password, **enable SSH**, and add a Wi-Fi. Your **phone's hotspot** is the best one: the robot then has internet for the AI wherever you go.
3. Write the card, put it in the Pi (slot underneath), power the Pi from the **power bank** (or a USB-C phone charger). Wait 2 minutes.
4. On the PC, double-click `pack_robot_for_pi.bat` in the project folder. It makes `raqib-robot.zip`.
5. Copy the zip to the Pi (WinSCP to `robot01.local`, or a USB stick) and run on the Pi:

```
unzip raqib-robot.zip -d ~/raqib
cd ~/raqib && bash edge/scripts/install_pi.sh
sudo reboot
```

The installer sets real-hardware mode, starts the robot and its server at every power-on, and makes the robot's own Wi-Fi **Raqib-Robot** (password `raqib1234`) when no known Wi-Fi is around.

Check before you continue

☐ The Pi boots and you can log in (SSH or keyboard + screen)

☐ The installer ended with 'Done'

☐ The phone app finds the robot ('Find automatically'); Robot health shows the sensors red: correct, nothing is wired yet

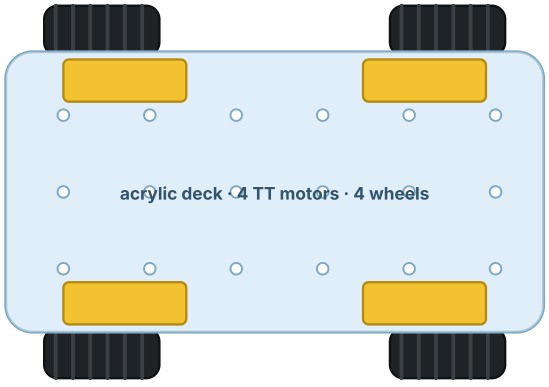
2 Chassis and 4 motors

You need: 4WD chassis kit (4 TT motors, 4 wheels), wire, soldering iron

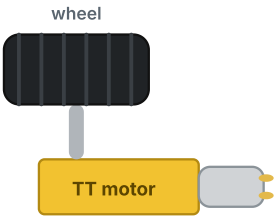
1. **Solder** a red and a black 20 cm wire to the two tabs of each of the 4 motors. Twist each pair.
2. Fix the 4 motors under the bottom deck with the kit's brackets and long screws, all wires towards the middle.
3. Push the 4 wheels onto the outer shafts.
4. Join the two motors of each side: red to red, black to black. Each side now has one red + one black wire.

How 4 wheels steer (skid steering): the two left wheels always turn together, and so do the two right wheels. Both sides forward = straight; one side slower = curve; sides in opposite directions = turn on the spot (the wheels skid a little sideways, which is normal). This is why one L298N is enough: one channel per side.

No wheel encoders in this build. The robot works out how far it drove from how long and how hard each motor ran (you calibrate this once in step 10), and the MPU6050 measures every turn. Fewer wires, fewer parts that can fail.



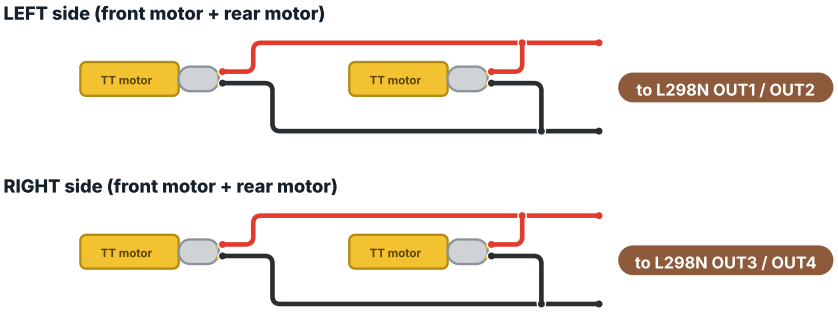
Screw the 4 motors under the bottom deck with the kit's brackets



TT motor + wheel (×4)

solder a red + black wire to each motor first

Four motors under the bottom deck.



Join red to red and black to black: the two motors of a side then work as one. A side that turns the wrong way: swap its two wires at the L298N.

Each side's two motors in parallel.

Check before you continue

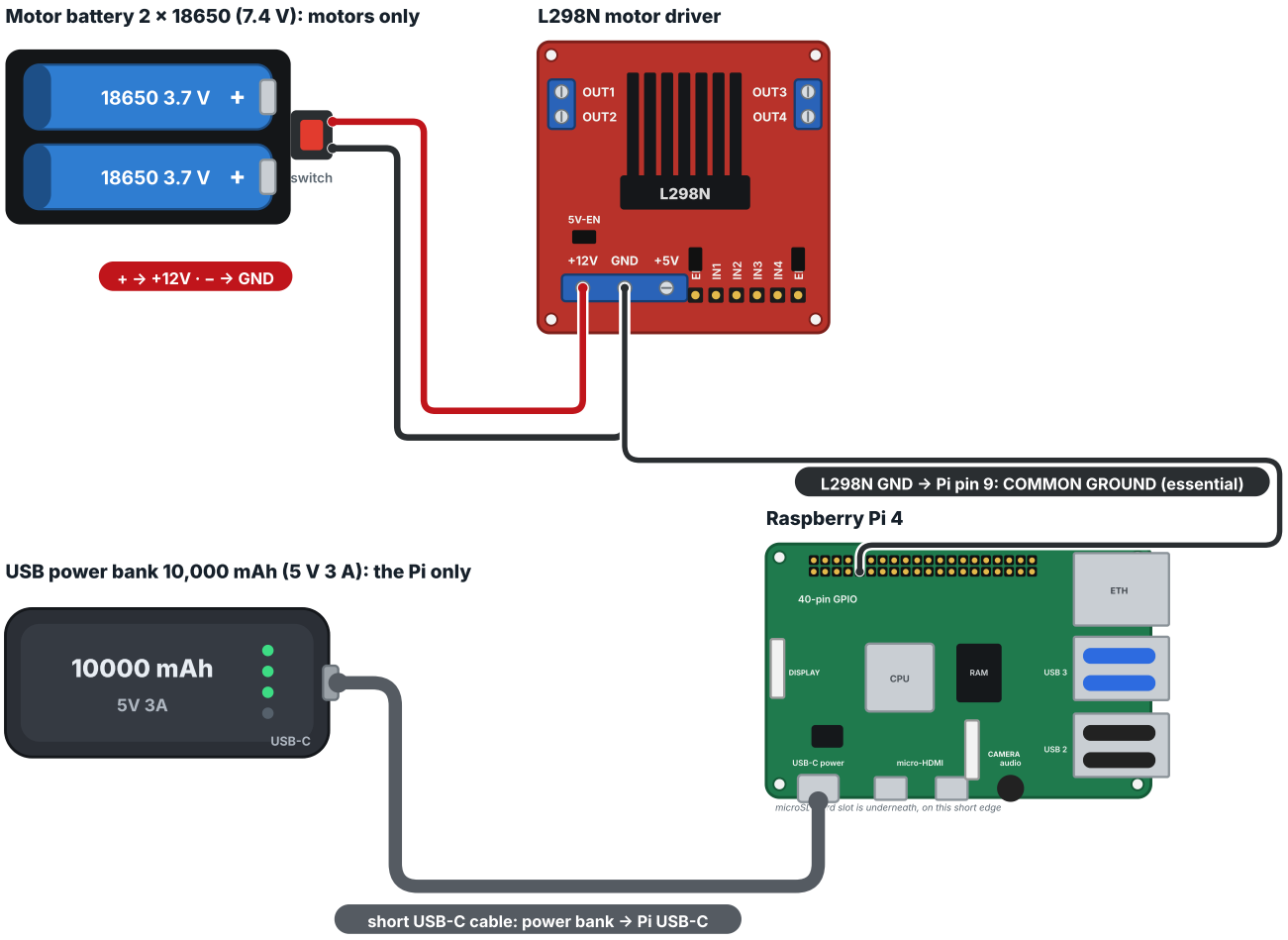
- ☐ All 4 wheels turn freely by hand
- ☐ Nothing rubs the wheels
- ☐ Left pair and right pair each have one red + one black wire to go to the L298N

3 Power: two batteries and one shared ground

You need: power bank + its USB-C cable, battery holder + 2 cells, L298N, wire, multimeter

The Pi and the motors get **separate batteries**. Motor starts pull a lot of current; on their own battery they can never make the Pi restart. There is nothing to adjust: the power bank already gives a clean 5 V.

1. Motor battery holder **red** → L298N **+12V**, **black** → L298N **GND**. Switch it off for now.
2. A second wire from the L298N **GND** screw → Raspberry Pi **pin 9** (GND). **Do not skip this:** without a shared ground the motor driver cannot understand the Pi's signals and the wheels do nothing or twitch.
3. Power bank → its USB-C cable → the Pi's **USB-C** power socket. The power bank sits on the bottom deck, the motor battery underneath.
4. **Never** connect the motor battery's + to the Pi or to the hub board.



Two batteries: motor current spikes can never restart the Pi, and there is no voltage to set by hand.

The one wire that joins the two systems is the common ground; without it the motors ignore the Pi.

Figure 3 - power wiring: the power bank feeds the Pi, the 18650 pack feeds the motors, one shared ground.

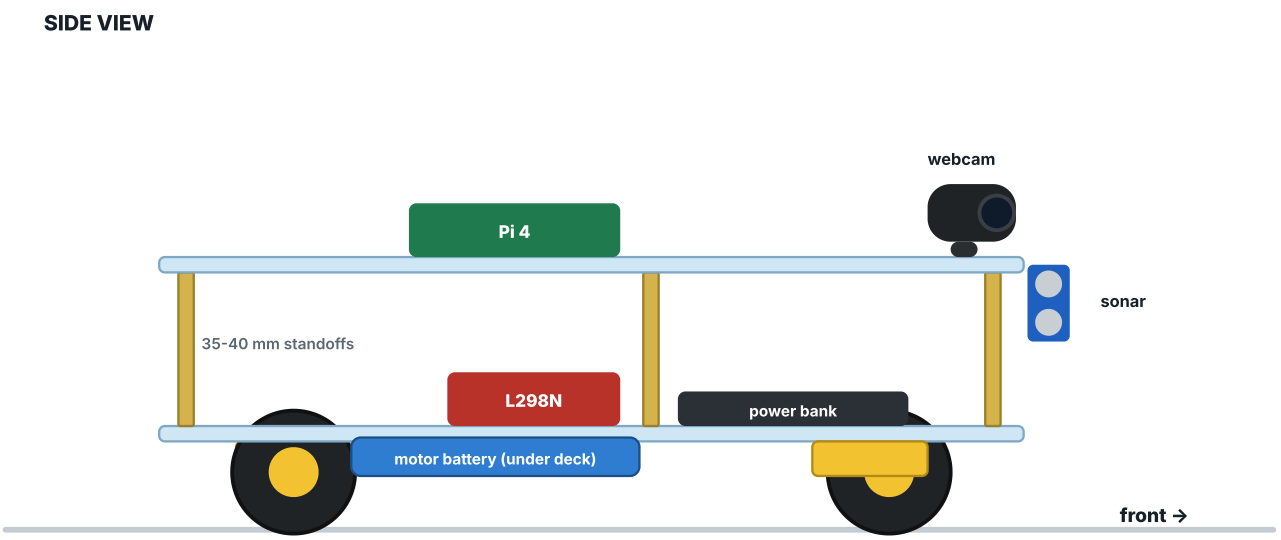
Check before you continue

- ☐ Motor battery reads 7-8.4 V at the L298N +12V terminal (switch on)
- ☐ The Pi boots from the power bank alone
- ☐ Continuity mode: L298N GND beeps to Pi pin 9
- ☐ Charging: power bank with its USB cable; motor cells in the 2-slot charger

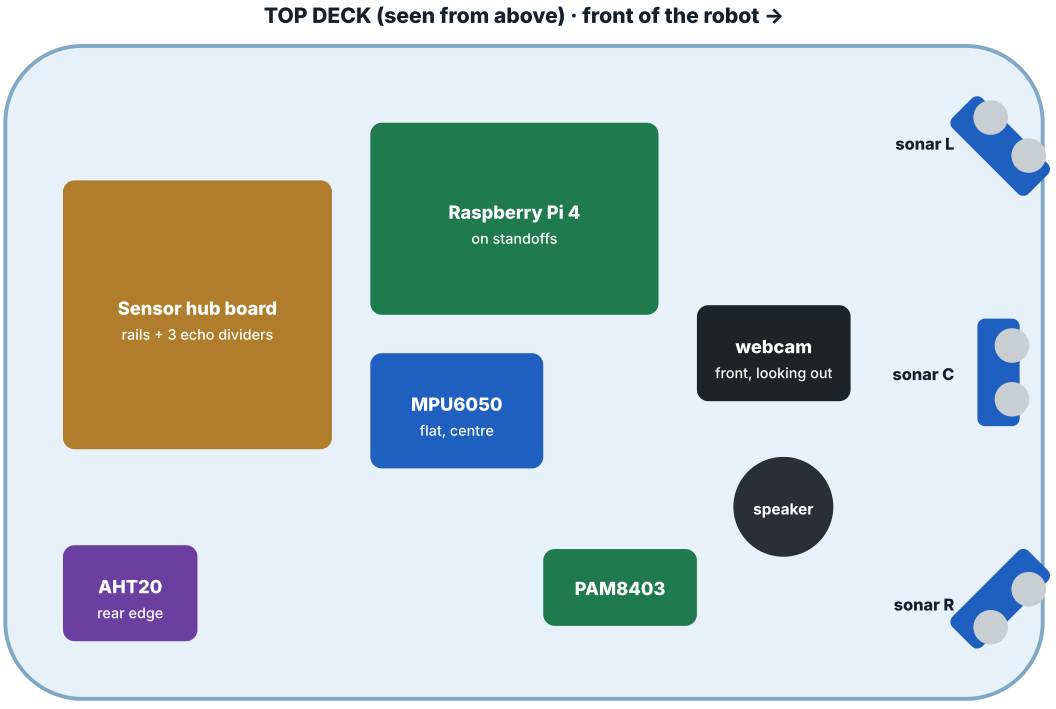
4 Mount the boards

You need: the kit's top deck and standoffs, screws, foam tape

Follow the layout: the motor battery under the bottom deck, the power bank and L298N on the bottom deck, the Pi on the top deck away from the AHT20, sonars on the front edge, the webcam at the front looking forward.



Side view: two decks on 35-40 mm standoffs.



Bottom deck: power bank and L298N; the 4 motors and the motor battery underneath. Keep the AHT20 away from the Pi CPU.

Top deck layout.

Check before you continue

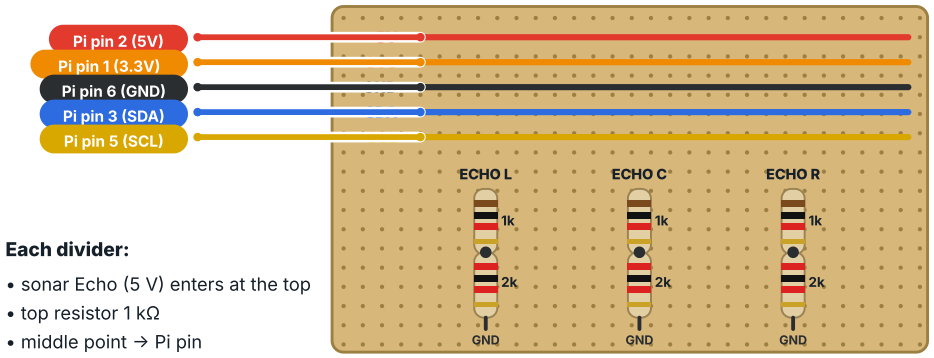
- ☐ Nothing touches the wheels
- ☐ No board's underside touches metal (use nylon washers)
- ☐ The power bank's button and USB port can still be reached to charge it

5 Build the hub board

You need: perfboard, resistors (3 × 1k, 3 × 2k), header strips, wire, soldering iron

The hub board is the heart of the wiring: five rails that every sensor plugs into, and the three sonar dividers.

1. Solder five long rails (bare wire or the board's strips): **5V, 3V3, GND, SDA, SCL**. Label them with a marker.
2. Solder a female header strip along each rail so jumper wires can plug in.
3. Solder the three sonar dividers: 1 kΩ on top, 2 kΩ to GND. Put a header pin on the input and on the middle point of each.
4. Connect the rails to the Pi with F-F jumpers: pin 2 → 5V, pin 1 → 3V3, pin 6 → GND, pin 3 → SDA, pin 5 → SCL.



Each divider:

- sonar Echo (5 V) enters at the top
- top resistor 1 kΩ
- middle point → Pi pin
- bottom resistor 2 kΩ → GND
- 5 V × 2/3 = 3.3 V: safe

Solder the rails as bare wire along the strips;
every module plugs into the rails with jumper wires.

Figure 4 - the hub board: rails on top, the three echo dividers below.

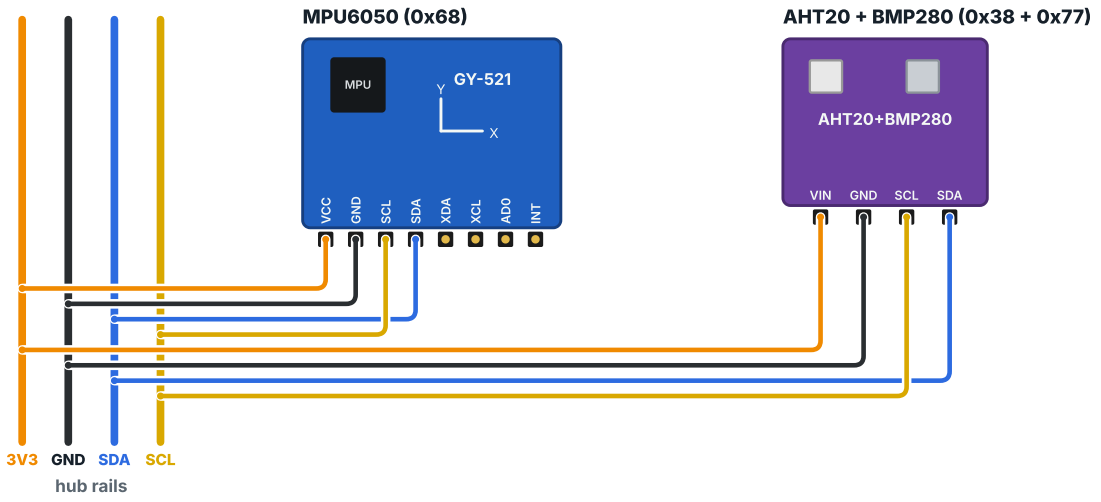
Check before you continue

- ☐ Multimeter, Pi on, nothing else plugged in: 5V rail ≈ 5 V, 3V3 rail ≈ 3.3 V
- ☐ No two rails are shorted (continuity mode: no beep between rails)
- ☐ Each divider: 5 V on its input reads ≈ 3.3 V at its middle point

6 I²C boards: MPU6050 and AHT20 + BMP280

You need: the two boards, F-F jumpers

Each I²C board gets four wires from the hub rails: **3V3, GND, SDA, SCL**.



- Both boards join the same four rails: 3.3 V, GND, SDA, SCL (Pi pins 1, 6, 3, 5).
- Each answers to its own address: MPU6050 0x68, AHT20 0x38, BMP280 0x77 (or 0x76).
- Test: python -m hwtest.i2c_scan lists every address found.

Figure 5 - the I²C bus.

```
sudo systemctl stop inspection-robot     # stop the robot program while you test
cd ~/raqib/edge && source .venv/bin/activate
python -m hwtest.i2c_scan
```

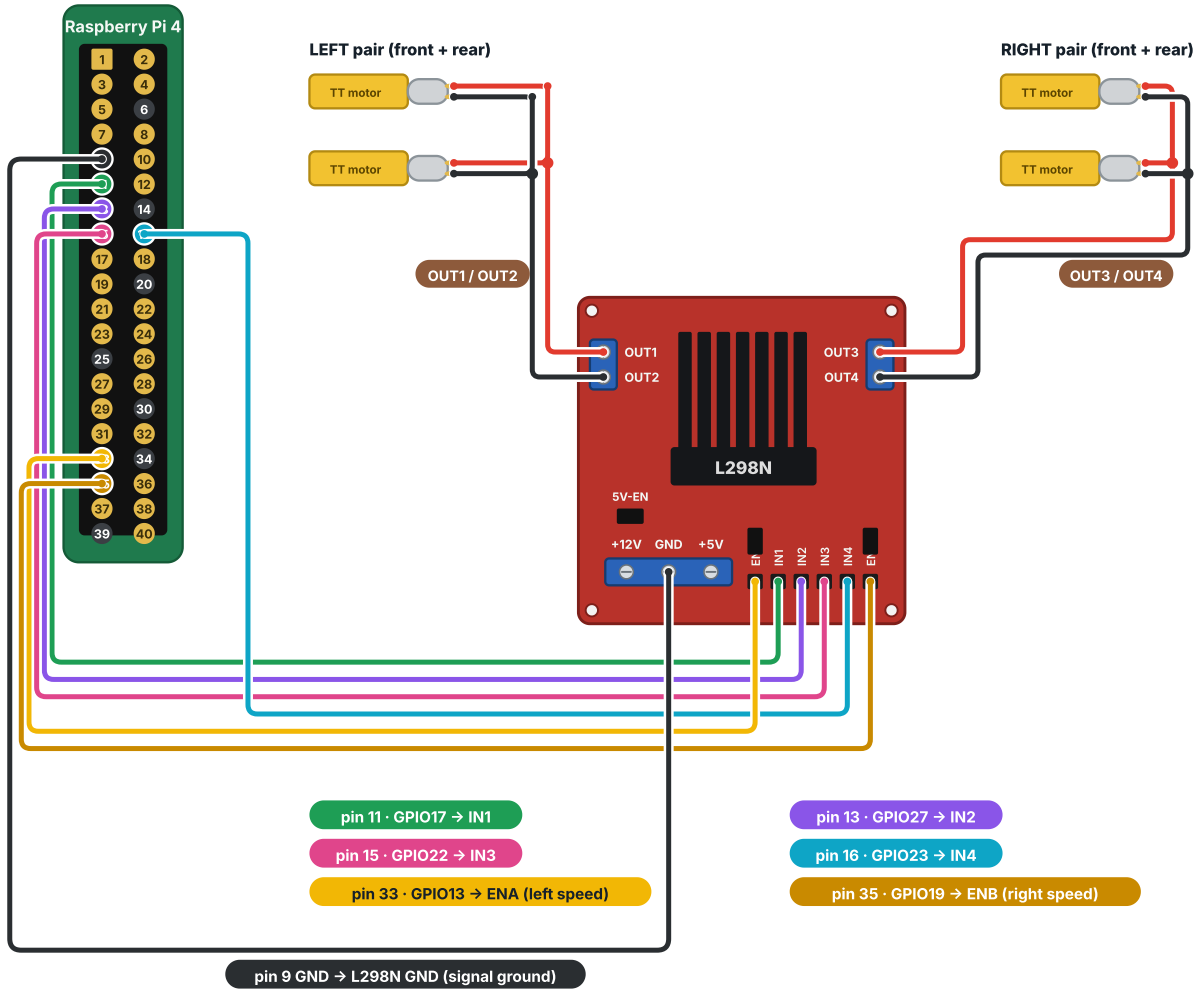
Check before you continue

- ☐ i2c_scan shows 0x38 + 0x77 or 0x76 (AHT20 + BMP280) and 0x68 (MPU6050)
- ☐ It says motor speed will come from GPIO 13/19 (no PCA9685 found): correct for this build

7 Motors: the L298N and the Pi

You need: L298N (already powered and grounded in step 3), M-F / F-F jumpers

1. Pull the **ENA** and **ENB** jumpers off the L298N. Leave the 5V-EN jumper.
2. Pi pins 11, 13, 15, 16 → L298N IN1, IN2, IN3, IN4 (direction).
3. Pi pin 33 → ENA, pin 35 → ENB (speed).
4. Left motor pair → OUT1/OUT2, right motor pair → OUT3/OUT4.
5. Check again: L298N GND → Pi pin 9 (step 3).



- Pull the ENA and ENB jumpers off: the Pi sets the speed on pins 33 and 35. Keep the 5V-EN jumper.
- +12V and GND come from the battery (power page). Leave the L298N's +5V terminal unconnected.
- Each side's two motors are joined (red to red, black to black) and run together: skid steering.

Figure 6 - motor wiring.

```
python -m hwtest.test_motors          # robot on a box, wheels in the air!
```

Check before you continue

☐ Each side turns forwards when told 'forward'. Wrong way: swap that pair's OUT wires (or invert: true in config)

☐ The wheels start smoothly (soft start), and stop at once

☐ The Pi keeps running when all 4 wheels start (it has its own battery)

8 Three distance sensors

You need: 3 HC-SR04, F-F jumpers, hot glue or brackets

1. Mount the sensors on the front edge: left one turned 45° left, centre straight, right one 45° right.
2. VCC → hub 5V, Gnd → hub GND.
3. Trig → Pi pin directly (3.3 V is enough to trigger).
4. Echo → its divider's input on the hub board; the divider's middle point → the Pi pin.

The software pings one sensor at a time with a short quiet gap, so the three never hear each other's echoes.

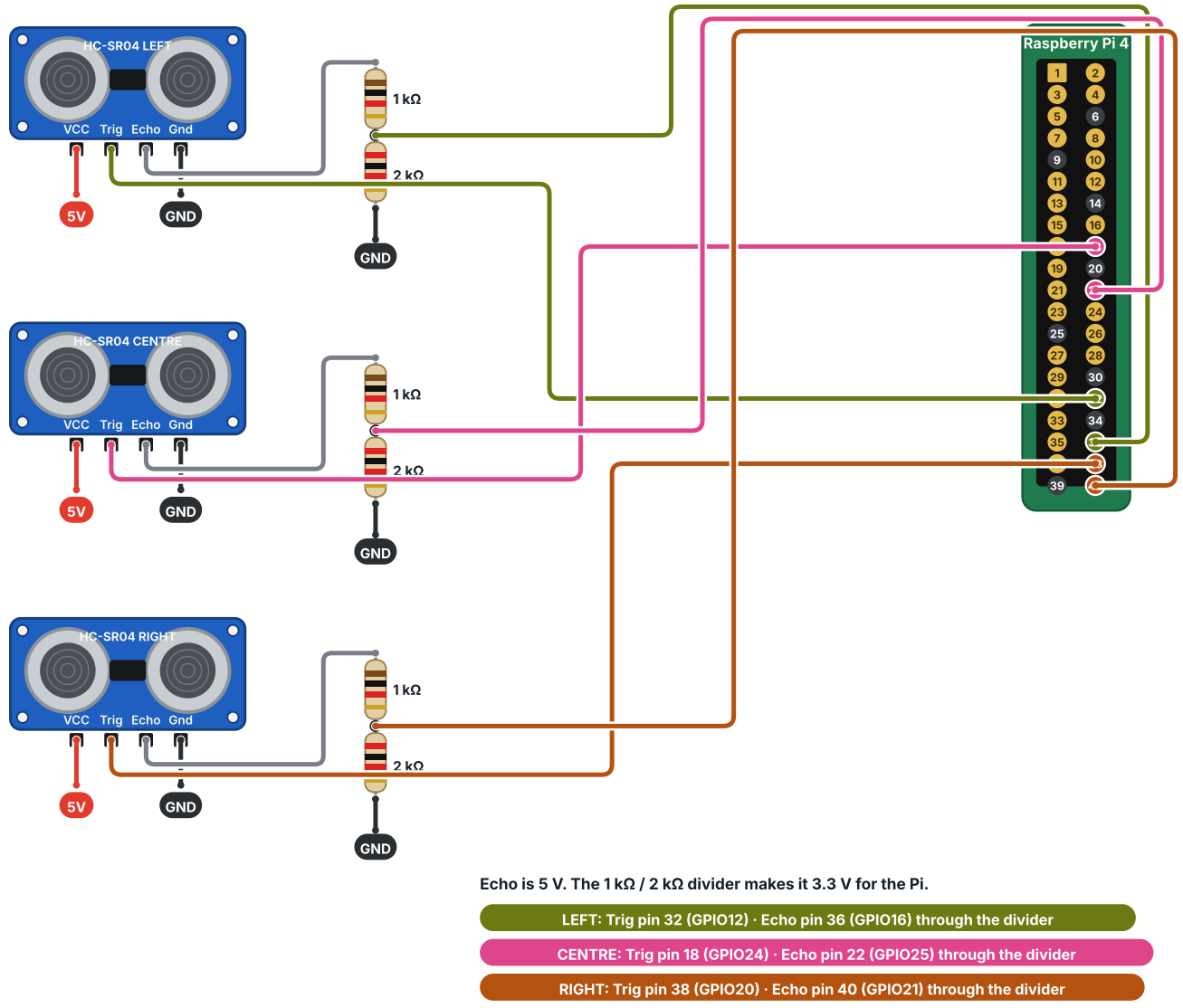


Figure 7 - ultrasonic sensors and their dividers.

```
python -m hwtest.test_sensors
```

Check before you continue

- ☐ Each distance follows your hand (left, centre, right)
- ☐ A constant 400 cm or 0 means Trig/Echo swapped or a divider fault
- ☐ Heading changes when you turn the robot by hand (MPU6050)

9 Webcam, microphone and speaker

You need: USB webcam, PAM8403, speaker, 3.5 mm cable

1. Webcam into a USB port; clip or glue it to the front of the top deck, level and looking forward.
2. Cut the 3.5 mm cable: the tip wire → PAM8403 L, the shield → G. Plug the other end into the Pi's audio jack. PAM8403 5V/GND from the hub.
3. Speaker → PAM8403 L+ and L−.

The robot finds the webcam, its microphone and the audio jack by itself.

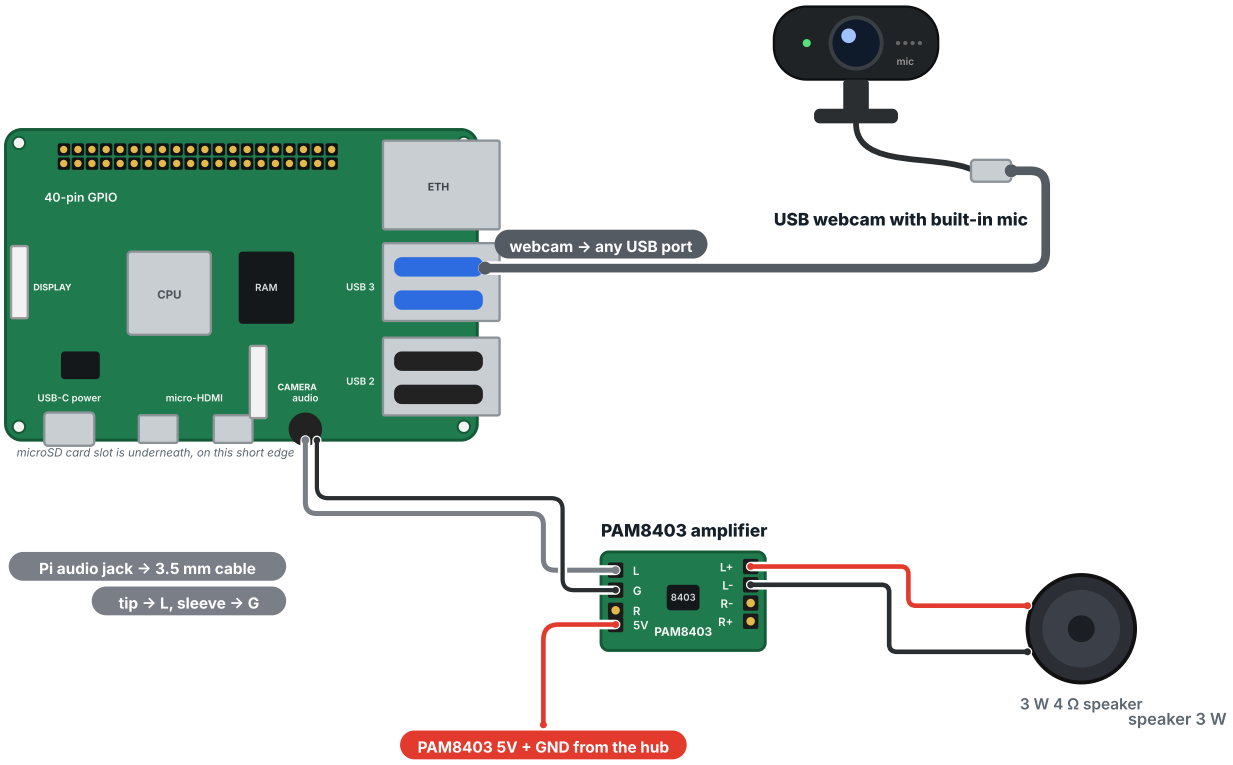


Figure 8 - webcam, audio jack, amplifier and speaker.

```
python -m hwtest.test_camera_audio
speaker-test -D plughw:CARD=Headphones -c1 -t wav -l1   # quick speaker check
```

Check before you continue

- ☐ test_snapshot.jpg shows the room the right way up
- ☐ test_audio.wav has your voice
- ☐ The speaker says the test sentence

10 Calibrate the driving (10 minutes, once)

You need: the robot, a tape measure, 3 m of free floor

Without encoders the robot measures distance from motor power × time, so it must learn how fast *your* motors are. Do this on the floor where the robot will work (tiles and carpet are different).

1. **Lowest power that moves:** robot on a box, wheels in the air. Run the first command and answer *y* as soon as all 4 wheels turn. Write the two numbers it prints into `edge/config.yaml` (`motor_deadband` and `min_power`).
2. **Top speed:** robot on the floor, mark the start with tape. Run the second command; it drives for 3 seconds. Measure how far it went in cm and type it in. Write the `max_speed_mps` it prints into `edge/config.yaml`.
3. **Check:** the third command drives 2 m. Measure: within ±10 % (1.8-2.2 m) is good. Not good? Charge the motor cells fully and repeat the top-speed run.

```
python -m hwtest.test_motors --deadband     # wheels in the air
python -m hwtest.test_motors --calibrate    # on the floor
python -m hwtest.test_motors --straight 2   # on the floor: check
```

Tip: calibrate with the motor cells fully charged, and charge them before each demo. Weak cells make the robot drive shorter than it thinks. Turning does not need calibrating: the MPU6050 measures every turn.

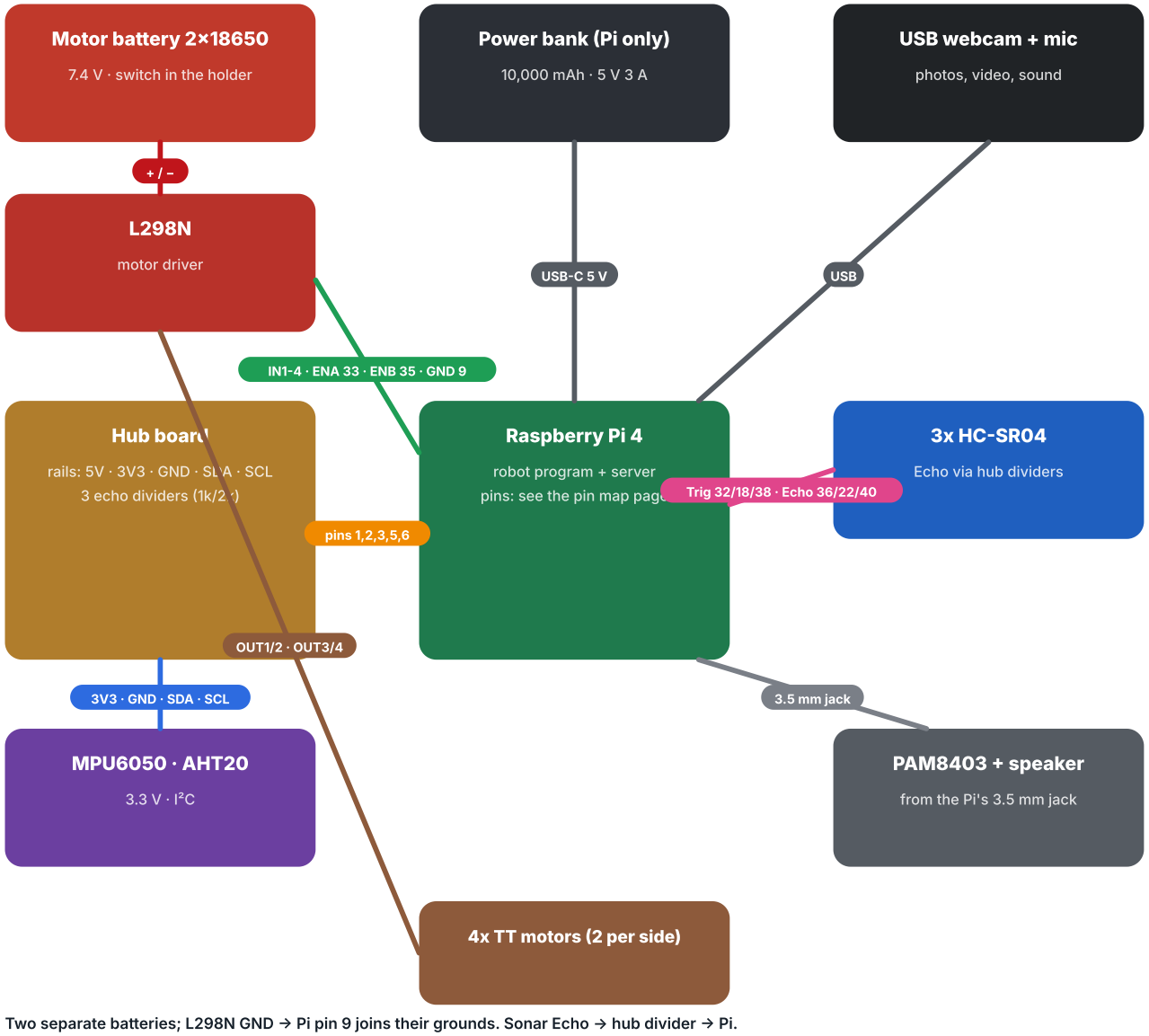
Check before you continue

- ☐ motor_deadband, min_power and max_speed_mps are written in config.yaml
- ☐ --straight 2 ends between 1.8 m and 2.2 m
- ☐ It drifts sideways less than 20 cm

11 Everything together

You need: the finished robot, the phone with the Raqib app

1. Tidy the cables with zip ties; nothing near the wheels.
2. Start the robot program again: `sudo systemctl start inspection-robot` (or reboot).
3. Phone on the same Wi-Fi (or join **Raqib-Robot**). Open the app → **Find automatically** → sign in.
4. **More → AI settings**: paste your Gemini key once.
5. **More → Robot health**: every part should be green. *Camera mount* says *fixed camera*, *Distance measuring* says *not fitted* and *Battery level* says *not measured*: all correct for this build. Red? Tap it for the fix, then **Re-check parts**.
6. Drive to each room, read the position on the Home screen, and write them under `zones:` in `edge/config.yaml`.



Two separate batteries; L298N GND → Pi pin 9 joins their grounds. Sonar Echo → hub divider → Pi.

Figure 9 - the complete harness on one page.

Check before you continue

- ☐ Every part green in Robot health
- ☐ STOP in the app stops the robot immediately
- ☐ A hand 30 cm in front of the moving robot stops it before 15 cm
- ☐ Drive → Camera: left turns the whole robot left, centre turns it back

7 Troubleshooting

What you see	Usual cause	Fix
i2c_scan finds nothing	I²C off, SDA/SCL swapped, or no 3.3 V on the rail	sudo raspi-config → Interface → I2C on; check pins 3/5; measure the 3V3 rail
One I²C board is missing	Loose jumper	Re-seat its 4 wires; run i2c_scan again
Wheels do nothing or twitch	No common ground between the L298N and the Pi	Wire L298N GND → Pi pin 9 (step 3)
Wheels do not turn	ENA/ENB jumpers still on, ENA/ENB not on pins 33/35, motor battery off	Remove the jumpers, check pins 33/35, switch on
One side turns backwards	Motor wires reversed	Swap that pair's two OUT wires, or set invert: true
One wheel of a pair turns the wrong way	That motor's red/black reversed	Swap its two wires where the pair is joined
Robot will not turn on the spot	4 wheels skid hard on carpet / rubber floor	Raise min_power to 0.45 and turn_speed to 0.6 in config.yaml
Drives too far or too short	Not calibrated on this floor, or weak motor cells	Charge the cells and run step 10 again
Lightning-bolt / 'low voltage' warning	Power bank gives less than 3 A, or a thin cable	Use a 5 V 3 A bank and its own short cable
Pi switches off during a run	Power bank empty	Charge it; its LEDs show the level
Sonar always 400 cm (or 0)	Trig/Echo swapped, no 5 V, wrong divider	Check the pin numbers and measure 5 V at the sensor; try the spare
Camera not found	Webcam not detected	ls /dev/video* should list it; try another USB port or cable
No sound	No amp power, cable on the wrong pins	5 V on the PAM8403; speaker-test (step 9); tip → L, shield → G
AI does not answer	No internet on the robot's Wi-Fi	Use the phone's hotspot as the robot's Wi-Fi (step 1)
App cannot find the robot	Phone on another network	Same Wi-Fi or join Raqib-Robot; the robot says its address at start-up

Logs on the Pi: `journalctl -u inspection-robot -f` (robot) and `journalctl -u raqib-server -f` (server). The app's **Robot health** screen shows every part with a hint when it is red.

8 Final checklist before a demo

☐ Power bank full (all LEDs on) and motor cells charged; a spare charged pair in the bag

☐ Driving calibrated on this floor (step 10): --straight 2 ends within 1.8-2.2 m

☐ Every part green in Robot health

☐ STOP in the app stops the robot; Reset is needed to move again

☐ A hand 30 cm in front of the moving robot stops it before 15 cm

☐ Stopping the server (sudo systemctl stop raqib-server) puts the robot in SAFE HOLD within about 10 s

☐ The robot starts at the Home spot, facing the same direction as when you measured the rooms

☐ The admin password in the app has been changed from admin123

☐ AI key entered in More → AI settings; "Test" says it works

9 Upgrades later

When you have more budget, add any of these. You do not change the software: at start-up the robot checks which parts answer and uses them (servos if a PCA9685 is found, the ribbon camera before the webcam, a BME680 before the AHT20, a UPS battery gauge). The gas sensor and encoders need one line in `edge/config.yaml` (`gas_sensor: mq135` , `encoders: true`). Press **Re-check parts** in Robot health after fitting one.

You want	Add	Extra RM
Gas / CO2 level ADS1115 on the I²C rails, MQ-135 AO through the divider to A0; set gas_sensor: mq135. The app shows gas readings again.	MQ-135 + ADS1115 + 10k/20k resistors	≈ 20
Exact distances Discs on the front motors, D0 → pins 29 / 31, VCC 3.3 V; set encoders: true. Distance is then counted, not timed.	2 slot encoders (LM393) + 20-slot discs	≈ 8
Look around without turning PCA9685 on the I²C rails, its ch0/ch1 → ENA/ENB (instead of pins 33/35), servos on ch14/ch15, webcam on the bracket.	PCA9685 + 2 SG90 servos + pan/tilt bracket	≈ 40
Sharper photos Ribbon into the CAMERA port. The robot tries the ribbon camera first, then the webcam.	Pi Camera OV5647 (≈ RM 35) or v2 (≈ RM 120) + long ribbon	35-128
Air quality (VOC) Replaces the AHT20 on the same 4 wires.	BME680 board	≈ 45
Battery % in the app Pi on the UPS; its battery gauge joins the I²C rails and the app shows the battery level.	UPS module 3S + 3 more 18650 cells (instead of the power bank)	≈ 175
Smoother video Same wiring.	Raspberry Pi 4, 4 GB instead of 2 GB	≈ 50

Built it? Run the final checklist, then show it off.